

MEDIATOR PATTERN

Behavioural Design Pattern

Mediator Pattern

Imagine you're building an **airport system** where multiple airplanes need to coordinate for landing and takeoff. If each airplane communicates **directly with every other airplane** to check if the runway is clear, the system becomes extremely complicated as more planes arrive. Each pilot would need to keep track of all other planes, creating a chaotic mess of communication lines and dependencies.

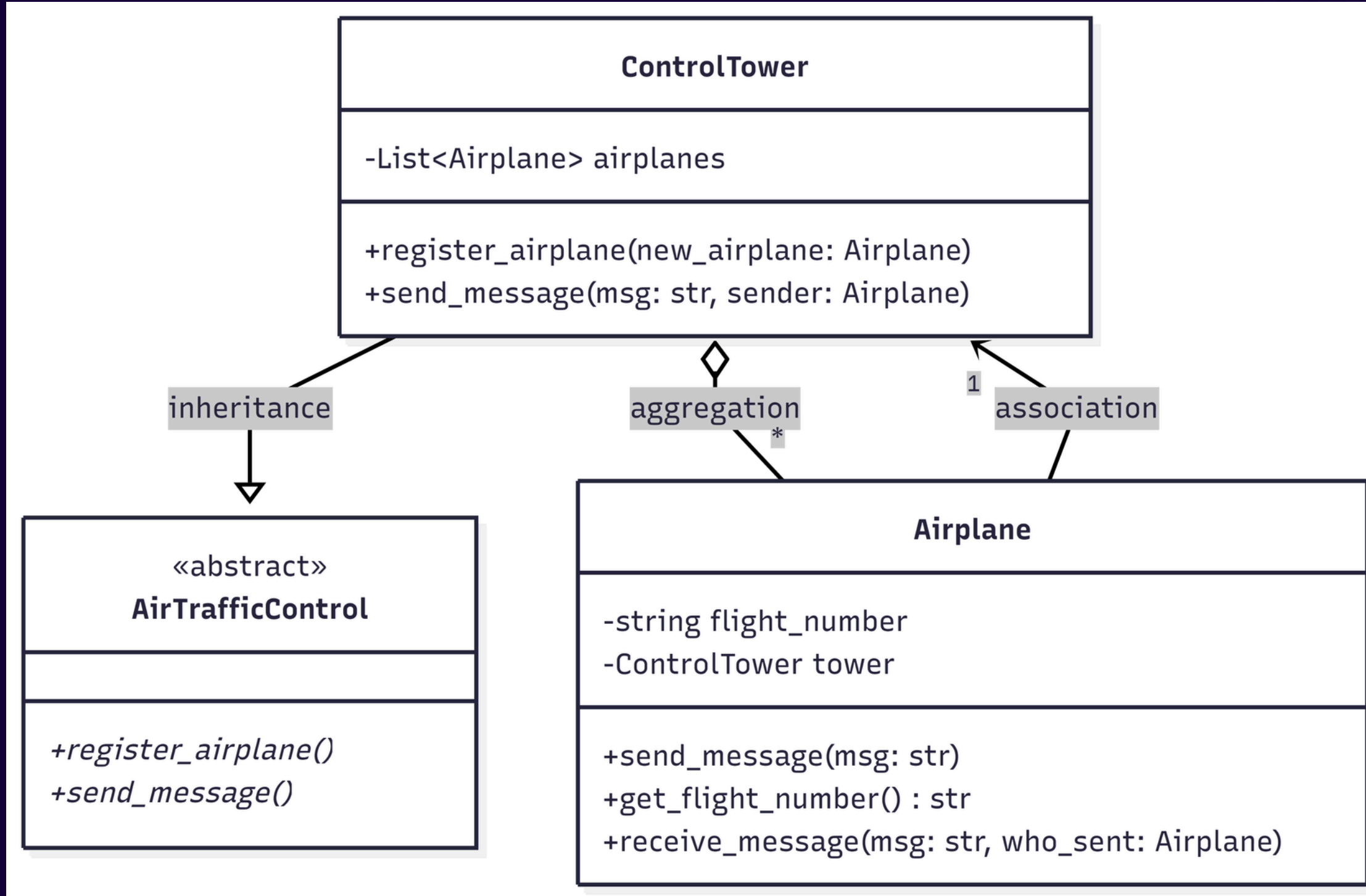
Solution using Mediator Pattern

Scenario: In an airport system, multiple airplanes communicate through a central **Air Traffic Control Tower**, eliminating the need for each plane to know about or track all other planes.

How it works:

- **Mediator:** Air Traffic Control Tower
- **Colleagues:** Airplanes send requests and messages to the control tower,
- which then coordinates and relays information to other planes as needed.

UML Diagram



Mediator Pattern Benefits

Reduces Complexity: The mediator handles all communication in one place, eliminating messy direct connections between objects.

Loose Coupling: Objects only interact with the mediator, making the code easier to modify and extend.

Single Responsibility: The mediator manages all communication logic, so individual objects can focus on their own tasks.

Centralized Control: Communication rules can be updated in the mediator without changing any of the individual objects.

Real World Use Cases

Air Traffic Control:

Airplanes communicate through a central control tower (mediator) instead of talking directly to each other.

GUI Component Coordination:

In user interface apps, different UI elements like dropdowns, text boxes, and buttons often need to work together. When you select something in a dropdown, it might need to update a text field or enable a button. A mediator handles these interactions so the components don't need to know about each other directly.

Workflow Systems:

In business management software, a mediator can coordinate tasks across different teams or departments, ensuring smooth communication without each team needing to directly interact with every other team.